Review Questions

The answers to the chapter review questions can be found in the Appendix.

1. Which code can be inserted to have the code print 2?

```
public class BirdSeed {
   private int numberBags;
   boolean call;

   public BirdSeed() {
      // LINE 1
      call = false;
      // LINE 2
   }

   public BirdSeed(int numberBags) {
      this.numberBags = numberBags;
   }

   public static void main(String[] args) {
      var seed = new BirdSeed();
      System.out.print(seed.numberBags);
   } }
```

   A. Replace line 1 with `BirdSeed(2);`.
   B. Replace line 2 with `BirdSeed(2);`.
   C. Replace line 1 with `new BirdSeed(2);`.
   D. Replace line 2 with `new BirdSeed(2);`.
   E. Replace line 1 with `this(2);`.
   F. Replace line 2 with `this(2);`.
   G. The code prints 2 without any changes.

2. Which modifier pairs can be used together in a method declaration? (Choose all that apply.)
   A. `static` and `final`
   B. `private` and `static`
   C. `static` and `abstract`
   D. `private` and `abstract`
   E. `abstract` and `final`
   F. `private` and `final`

3. Which of the following statements about methods are true? (Choose all that apply.)
   A. Overloaded methods must have the same signature.
   B. Overridden methods must have the same signature.
   C. Hidden methods must have the same signature.
   D. Overloaded methods must have the same return type.
   E. Overridden methods must have the same return type.
   F. Hidden methods must have the same return type.

4. What is the output of the following program?

```
1:  class Mammal {
2:     private void sneeze() {}
3:     public Mammal(int age) {
4:        System.out.print("Mammal");
5:     } }
6:  public class Platypus extends Mammal {
7:     int sneeze() { return 1; }
8:     public Platypus() {
9:        System.out.print("Platypus");
10:    }
11:    public static void main(String[] args) {
12:       new Mammal(5);
13:    } }
```

   A. `Platypus`
   B. `Mammal`
   C. `PlatypusMammal`
   D. `MammalPlatypus`
   E. The code will compile if line 7 is changed.
   F. The code will compile if line 9 is changed.

5. Which of the following complete the constructor so that this code prints out `50`? (Choose all that apply.)

```
class Speedster {
   int numSpots;
}
public class Cheetah extends Speedster {
   int numSpots;

   public Cheetah(int numSpots) {
      // INSERT CODE HERE
   }
```

```
    public static void main(String[] args) {
        Speedster s = new Cheetah(50);
        System.out.print(s.numSpots);
    }
}
```

A. `numSpots = numSpots;`

B. `numSpots = this.numSpots;`

C. `this.numSpots = numSpots;`

D. `numSpots = super.numSpots;`

E. `super.numSpots = numSpots;`

F. The code does not compile regardless of the code inserted into the constructor.

G. None of the above

6. Which of the following declare immutable classes? (Choose all that apply.)

```
public final class Moose {
    private final int antlers;
}

public class Caribou {
    private int antlers = 10;
}

public class Reindeer {
    private final int antlers = 5;
}

public final class Elk {}

public final class Deer {
    private final Object o = new Object();
}
```

A. `Moose`

B. `Caribou`

C. `Reindeer`

D. `Elk`

E. `Deer`

F. None of the above

7. What is the output of the following code?

```
1:  class Arthropod {
2:     protected void printName(long input) {
3:        System.out.print("Arthropod");
4:     }
5:     void printName(int input) {
6:        System.out.print("Spooky");
7:     } }
8:  public class Spider extends Arthropod {
9:     protected void printName(int input) {
10:       System.out.print("Spider");
11:    }
12:    public static void main(String[] args) {
13:       Arthropod a = new Spider();
14:       a.printName((short)4);
15:       a.printName(4);
16:       a.printName(5L);
17:    } }
```

A. `SpiderSpiderArthropod`

B. `SpiderSpiderSpider`

C. `SpiderSpookyArthropod`

D. `SpookySpiderArthropod`

E. The code will not compile because of line 5.

F. The code will not compile because of line 9.

G. None of the above

8. What is the result of the following code?

```
abstract class Bird {
   private final void fly() { System.out.println("Bird"); }
   protected Bird() { System.out.print("Wow-"); }
}
public class Pelican extends Bird {
   public Pelican() { System.out.print("Oh-"); }
   protected void fly() { System.out.println("Pelican"); }
   public static void main(String[] args) {
      var chirp = new Pelican();
      chirp.fly();
} }
```

- **A.** `Oh-Bird`
- **B.** `Oh-Pelican`
- **C.** `Wow-Oh-Bird`
- **D.** `Wow-Oh-Pelican`
- **E.** The code contains a compilation error.
- **F.** None of the above

9. Which of the following statements about overridden methods are true? (Choose all that apply.)

- **A.** An overridden method must contain method parameters that are the same or covariant with the method parameters in the inherited method.
- **B.** An overridden method may declare a new exception, provided it is not checked.
- **C.** An overridden method must be more accessible than the method in the parent class.
- **D.** An overridden method may declare a broader checked exception than the method in the parent class.
- **E.** If an inherited method returns `void`, then the overridden version of the method must return `void`.
- **F.** None of the above

10. Which of the following pairs, when inserted into the blanks, allow the code to compile? (Choose all that apply.)

```
1:  public class Howler {
2:     public Howler(long shadow) {
3:        _______________;
4:     }
5:     private Howler(int moon) {
6:        super();
7:     }
8:  }
9:  class Wolf extends Howler {
10:    protected Wolf(String stars) {
11:       super(2L);
12:    }
13:    public Wolf() {
14:       _______________;
15:    }
16: }
```

- **A.** `this(3)` at line 3, `this("")` at line 14
- **B.** `this()` at line 3, `super(1)` at line 14
- **C.** `this((short)1)` at line 3, `this(null)` at line 14
- **D.** `super()` at line 3, `super()` at line 14

E. `this(2L)` at line 3, `super((short)2)` at line 14

F. `this(5)` at line 3, `super(null)` at line 14

G. Remove lines 3 and 14.

11. What is the result of the following?

```
public class PolarBear {
   StringBuilder value = new StringBuilder("t");
   { value.append("a"); }
   { value.append("c"); }
   private PolarBear() {
      value.append("b");
   }
   public PolarBear(String s) {
      this();
      value.append(s);
   }
   public PolarBear(CharSequence p) {
      value.append(p);
   }
   public static void main(String[] args) {
      Object bear = new PolarBear();
      bear = new PolarBear("f");
      System.out.println(((PolarBear)bear).value);
   } }
```

A. `tacb`

B. `tacf`

C. `tacbf`

D. `tcafb`

E. `taftacb`

F. The code does not compile.

G. An exception is thrown.

12. How many lines of the following program contain a compilation error?

```
public class Rodent {
   public Rodent(Integer x) {}
   protected static Integer chew() throws Exception {
      System.out.println("Rodent is chewing");
      return 1;
   }
```

```
}
class Beaver extends Rodent {
   public Number chew() throws RuntimeException {
      System.out.println("Beaver is chewing on wood");
      return 2;
   } }
```

A. None

B. 1

C. 2

D. 3

E. 4

F. 5

13. Which of these classes compile and will include a default constructor created by the compiler? (Choose all that apply.)

A.

```
public class Bird {}
```

B.

```
public class Bird {
   public bird() {}
}
```

C.

```
public class Bird {
   public bird(String name) {}
}
```

D.

```
public class Bird {
   public Bird() {}
}
```

E.

```
public class Bird {
   Bird(String name) {}
}
```

F.

```
public class Bird {
   private Bird(int age) {}
}
```

G.

```
public class Bird {
   public Bird bird() { return null; }
}
```

14. Which of the following statements about inheritance are correct? (Choose all that apply.)

A. A class can directly extend any number of classes.

B. A class can implement any number of interfaces.

C. All variables inherit `java.lang.Object`.

D. If class `A` is extended by `B`, then `B` is a superclass of `A`.

E. If class `C` implements interface `D`, then `C` is a subtype of `D`.

F. Multiple inheritance is the property of a class to have multiple direct superclasses.

15. Which statements about the following program are correct? (Choose all that apply.)

```
1: abstract class Nocturnal {
2:    boolean isBlind();
3: }
4: public class Owl extends Nocturnal {
5:    public boolean isBlind() { return false; }
6:    public static void main(String[] args) {
7:       var nocturnal = (Nocturnal)new Owl();
8:       System.out.println(nocturnal.isBlind());
9: } }
```

A. It compiles and prints `true`.

B. It compiles and prints `false`.

C. The code will not compile because of line 2.

D. The code will not compile because of line 5.

E. The code will not compile because of line 7.

F. The code will not compile because of line 8.

G. None of the above

16. What is the result of the following?

```
class Arachnid {
   static StringBuilder sb = new StringBuilder();
   { sb.append("c"); }
   static
   { sb.append("u"); }
   { sb.append("r"); }
}
```

```
public class Scorpion extends Arachnid {
   static
   { sb.append("q"); }
   { sb.append("m"); }
   public static void main(String[] args) {
      System.out.print(Scorpion.sb + " ");
      System.out.print(Scorpion.sb + " ");
      new Arachnid();
      new Scorpion();
      System.out.print(Scorpion.sb);
   } }
```

- **A.** `qu qu qumrcrc`
- **B.** `u u ucrcrm`
- **C.** `uq uq uqmcrcr`
- **D.** `uq uq uqcrcrm`
- **E.** `qu qu qumcrcr`
- **F.** `qu qu qucrcrm`
- **G.** The code does not compile.

17. Which of the following are true? (Choose all that apply.)

- **A.** `this()` can be called from anywhere in a constructor.
- **B.** `this()` can be called from anywhere in an instance method.
- **C.** `this.variableName` can be called from any instance method in the class.
- **D.** `this.variableName` can be called from any `static` method in the class.
- **E.** You can call the default constructor written by the compiler using `this()`.
- **F.** You can access a `private` constructor with the `main()` method in the same class.

18. Which statements about the following classes are correct? (Choose all that apply.)

```
public class Mammal {
   private void eat() {}
   protected static void drink() {}
   public Integer dance(String p) { return null; }
}
class Primate extends Mammal {
   public void eat(String p) {}
}
class Monkey extends Primate {
```

```
10:    public static void drink() throws RuntimeException {}
11:    public Number dance(CharSequence p) { return null; }
12:    public int eat(String p) {}
13: }
```

A. The `eat()` method in `Mammal` is correctly overridden on line 7.

B. The `eat()` method in `Mammal` is correctly overloaded on line 7.

C. The `drink()` method in `Mammal` is correctly overridden on line 10.

D. The `drink()` method in `Mammal` is correctly hidden on line 10.

E. The `dance()` method in `Mammal` is correctly overridden on line 11.

F. The `dance()` method in `Mammal` is correctly overloaded on line 11.

G. The `eat()` method in `Primate` is correctly hidden on line 12.

H. The `eat()` method in `Primate` is correctly overloaded on line 12.

19. What is the output of the following code?

```
1:  class Reptile {
2:     {System.out.print("A");}
3:     public Reptile(int hatch) {}
4:     void layEggs() {
5:        System.out.print("Reptile");
6:     } }
7:  public class Lizard extends Reptile {
8:     static {System.out.print("B");}
9:     public Lizard(int hatch) {}
10:    public final void layEggs() {
11:       System.out.print("Lizard");
12:    }
13:    public static void main(String[] args) {
14:       var reptile = new Lizard(1);
15:       reptile.layEggs();
16:    } }
```

A. `AALizard`

B. `BALizard`

C. `BLizardA`

D. `ALizard`

E. The code will not compile because of line 3.

F. None of the above

20. Which statement about the following program is correct?

```
class Bird {
   int feathers = 0;
   Bird(int x) { this.feathers = x; }
   Bird fly() {
      return new Bird(1);
   } }
class Parrot extends Bird {
   protected Parrot(int y) { super(y); }
   protected Parrot fly() {
      return new Parrot(2);
   } }
public class Macaw extends Parrot {
   public Macaw(int z) { super(z); }
   public Macaw fly() {
      return new Macaw(3);
   }
   public static void main(String... sing) {
      Bird p = new Macaw(4);
      System.out.print(((Parrot)p.fly()).feathers);
   } }
```

A. One line contains a compiler error.
B. Two lines contain compiler errors.
C. Three lines contain compiler errors.
D. The code compiles but throws a `ClassCastException` at runtime.
E. The program compiles and prints `3`.
F. The program compiles and prints `0`.

21. Which of the following are properties of immutable classes? (Choose all that apply.)

A. The class can contain setter methods, provided they are marked `final`.
B. The class must not be able to be extended outside the class declaration.
C. The class may not contain any instance variables.
D. The class must be marked `static`.
E. The class may not contain any `static` variables.
F. The class may only contain `private` constructors.
G. The data for mutable instance variables may be read, provided they cannot be modified by the caller.

22. What does the following program print?

```
class Person {
   static String name;
   void setName(String q) { name = q; } }
public class Child extends Person {
   static String name;
   void setName(String w) { name = w; }
   public static void main(String[] p) {
      final Child m = new Child();
      final Person t = m;
      m.name = "Elysia";
      t.name = "Sophia";
      m.setName("Webby");
      t.setName("Olivia");
      System.out.println(m.name + " " + t.name);
   } }
```

A. `Elysia Sophia`

B. `Webby Olivia`

C. `Olivia Olivia`

D. `Olivia Sophia`

E. The code does not compile.

F. None of the above

23. What is the output of the following program?

```
class Canine {
   public Canine(boolean t) { logger.append("a"); }
   public Canine() { logger.append("q"); }
   private StringBuilder logger = new StringBuilder();
   protected void print(String v) { logger.append(v); }
   protected String view() { return logger.toString(); }
}
class Fox extends Canine {
   public Fox(long x) { print("p"); }
   public Fox(String name) {
      this(2);
      print("z");
   }
```

```
}

public class Fennec extends Fox {
   public Fennec(int e) {
      super("tails");
      print("j");
   }
   public Fennec(short f) {
      super("eevee");
      print("m");
   }

   public static void main(String... unused) {
      System.out.println(new Fennec(1).view());
   } }
```

A. `qpz`

B. `qpzj`

C. `jzpa`

D. `apj`

E. `apjm`

F. The code does not compile.

G. None of the above

24. What is printed by the following program?

```
class Antelope {
   public Antelope(int p) {
      System.out.print("4");
   }
   { System.out.print("2"); }
   static { System.out.print("1"); }
}
public class Gazelle extends Antelope {
   public Gazelle(int p) {
      super(6);
      System.out.print("3");
   }
```

```
   public static void main(String hopping[]) {
      new Gazelle(0);
   }
   static { System.out.print("8"); }
   { System.out.print("9"); }
}
```

- **A.** `182640`
- **B.** `182943`
- **C.** `182493`
- **D.** `421389`
- **E.** The code does not compile.
- **F.** The output cannot be determined until runtime.

25. Which of the following are true about a concrete class? (Choose all that apply.)

- **A.** A concrete class can be declared as `abstract`.
- **B.** A concrete class must implement all inherited abstract methods.
- **C.** A concrete class can be marked as `final`.
- **D.** A concrete class must be immutable.
- **E.** A concrete method that implements an abstract method must match the method declaration of the abstract method exactly.

26. What is the output of the following code?

```
public abstract class Whale {
   public abstract void dive();
   public static void main(String[] args) {
      Whale whale = new Orca();
      whale.dive(3);
   }
}
class Orca extends Whale {
   static public int MAX = 3;
   public void dive() {
      System.out.println("Orca diving");
   }
   public void dive(int... depth) {
      System.out.println("Orca diving deeper "+MAX);
} }
```

A. `Orca diving`

B. `Orca diving deeper 3`

C. The code will not compile because of line 4.

D. The code will not compile because of line 8.

E. The code will not compile because of line 11.

F. The code will not compile because of line 12.

G. The code will not compile because of line 17.

H. None of the above